Text Embedding Models - Technical Report

Text embedding models convert natural language into dense numerical vectors that capture semantic meaning. Similar texts map to nearby points in the vector space, which enables semantic search, clustering, recommendation, and retrieval augmented generation.

Modern embedding models are trained with contrastive learning: the model learns to pull semantically related pairs (a query and its relevant passage) close together while pushing unrelated pairs apart. Training typically uses a unified multitask learning framework that combines retrieval, classification, and semantic similarity objectives.

Hard Negative Mining Techniques

Hard negative mining is a technique for improving embedding model quality. Instead of sampling random negatives, hard negatives are passages that are lexically or semantically similar to the query but do not actually answer it. Training against several hard-candidate negatives forces the model to learn fine-grained distinctions and substantially improves benchmark ranking performance.

Common strategies include mining negatives from a first-stage BM25 retriever, using a previously trained dense retriever to select high-scoring but incorrect passages, and periodically refreshing the negative pool during training as the model improves.

Evaluation and Dimensionality

Embedding models are evaluated on benchmarks such as MTEB (Massive Text Embedding Benchmark) covering retrieval, reranking, clustering, and semantic textual similarity tasks.

The all-MiniLM-L6-v2 sentence-transformers model is a widely used open-source embedding model that produces 384-dimensional vectors. It offers an excellent trade-off between speed and quality, which makes it a popular default choice for RAG prototypes and production systems with tight latency budgets.